The two membership heads are independently necessary for their own derivability in the PA196 calculus

Research note

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Abstract

A tempting state reduction deletes one of PA196's two membership rewriting heads and tries to recover it using the other and pointed separation. Two explicit interpretations show that this particular strategy cannot work: each head fails in an interpretation validating all the remaining exact native rules. The interpretation is sensitive to the numerical labels of variables, which is permitted when proving independence of this deliberately restricted numeric calculus. We also give a nontrivial consequence of one-sided rewriting, namely ordinary same-variable universal elimination, to distinguish the obstruction from the failure of the simplest specialization arguments.

1 The question

The exact definitions are those of `PA196/Logic/Basic.lean`. Up to exchange of antecedents, the two heads are

$$\begin{aligned} L(a,b,d): &\quad \mathsf{E}(a,b) \to (\mathsf{M}(a,d) \to \mathsf{M}(b,d)),\\ R(a,b,d): &\quad \mathsf{E}(a,b) \to (\mathsf{M}(d,a) \to \mathsf{M}(d,b)). \end{aligned}$$

They are the paper's head 3 and head 4, respectively. The remaining rules are modus ponens, closed generalization, heads 1, 2, 5 and 6, the restricted propositional head ρ_8, and the entire guarded gate, including equality vacuity and membership vacuity.

An independence interpretation need only validate these exact numeric rules. It need not validate all unpostulated rules of ordinary first-order logic, especially uniform substitution of variable labels inside membership atoms. Assuming that additional invariance would assume precisely the phenomenon to be derived.

2 A two-point permutation

Let $\sigma : \mathbb{N} \to \mathbb{N}$ interchange 2 and 3 and fix all other numbers. It is an involution. Both identities

$$\neg\,\mathrm{Bit}(\sigma(t), t), \qquad \neg\,\mathrm{Bit}(t, \sigma(t))$$

hold for every t. Away from $2, 3$ they are the usual absence of the tth bit in t. At $2, 3$ they follow by direct calculation: $\mathrm{Bit}(3, 2) = \mathrm{Bit}(2, 3) = 0$. Set $f_1 = g_1 = \sigma$ and $f_i = g_i = \mathrm{id}$ for $i \neq 1$.

3 Head 3 is not derivable without itself

Given an assignment $\nu : \mathbb{N} \to \mathbb{N}$, interpret atoms by

$$\mathsf{M}(a,b)^{\nu} := \mathrm{Bit}\big(f_a(\nu(a)), \nu(b)\big), \qquad \mathsf{E}(a,b)^{\nu} := (\nu(a) = \nu(b)).$$

Interpret implication classically, and interpret $\mathsf{U}(a, B)$ as universal quantification of the ath coordinate of the assignment. Use the exact numeric decoder, including code 0 as truth and the reversed quantifier pairing.

Since $f_0 = \mathrm{id}$, code $1 = \mathsf{M}(0,0)$ is false under every assignment. Hence all the derived connectives have their intended classical meanings, even when quantification binds coordinate 0.

The quantifier heads 1, 2 and 6, the equality witness head 5, and ρ_8 are valid for ordinary reasons. Head 4 is valid: if $\nu(a) = \nu(b)$, the two membership formulas in $R(a,b,d)$ test the same digit of the same number $f_d(\nu(d))$. Modus ponens and closed generalization preserve validity.

It remains to check the entire gate, because arbitrary separation is the possible source of a hidden derivation. Let $c \neq a, b$ and let φ be the decoded property with original code c. The source-label bound shows that changing coordinate c does not change φ. Write

$$P(y) := \varphi^{\nu[b:=y]}.$$

If $a \neq b$, take the finite set

$$Z = \{y : (\mathrm{Bit}(f_a(\nu(a)), y) \vee y = \nu(b)) \wedge P(y)\}.$$

Let $m = \sum_{y \in Z} 2^y$, and choose $s = f_c^{-1}(m)$. Then $\mathsf{M}(c,b)$ under $\nu[c := s, b := y]$ is precisely $y \in Z$. This establishes the separation conjunct and the distinguished-point clause. If $a = b$, the first antecedent of the separation conjunct is $\mathrm{Bit}(f_b(y), y)$, which is always false. In this case use just $Z = \{\nu(b) : P(\nu(b))\}$; the remaining subset condition and the point clause still hold. Equality and membership vacuity are valid because $c \neq a, b$. Thus the whole bundled gate is valid.

But head 3 fails. Use the actual instance

$$\mathsf{M}(1,2) \to \big(\mathsf{E}(1,0) \to \mathsf{M}(0,2)\big)$$

and the assignment $\nu(1) = \nu(0) = 2$, $\nu(2) = 0$. Its first antecedent is $\mathrm{Bit}(3,0)$, which is true; its equality antecedent is true; its conclusion is $\mathrm{Bit}(2,0)$, which is false.

Theorem 1. *Head 3 is not derivable from the remaining exact PA196 native rules. This remains true if primitive equality reflexivity, symmetry and transitivity are added, since equality in the counterinterpretation is actual equality of natural numbers.*

Proof. Every remaining rule preserves validity in the interpretation, and the displayed instance is invalid. □

4 Head 4 is not derivable without itself

Use instead

$$\mathsf{M}(a,b)^\nu := \mathrm{Bit}\big(\nu(a), g_b(\nu(b))\big), \qquad \mathsf{E}(a,b)^\nu := (\nu(a) = \nu(b)).$$

The same argument verifies the logical heads and rules. Head 3 is now valid, because it replaces the container value by an equal value and leaves the digit index $g_d(\nu(d))$ unchanged.

For the gate with $a \neq b$, consider the finite set of indices

$$Z = \{g_b(y) : (\mathrm{Bit}(\nu(a), g_b(y)) \vee y = \nu(b)) \wedge \varphi^{\nu[b:=y]}\}.$$

It is finite because g_b is a bijection and a natural number has only finitely many set bits. Take the witness $s = \sum_{z \in Z} 2^z$ for coordinate c. Injectivity of g_b ensures that no point with the wrong property enters the selected set. If $a = b$, the first antecedent is $\mathrm{Bit}(y, g_b(y))$, always false; the singleton-point construction suffices again. All other parts of the gate are as before.

Head 4 fails at

$$\mathsf{E}(0,1) \to \big(\mathsf{M}(2,0) \to \mathsf{M}(2,1)\big)$$

with $\nu(0) = \nu(1) = 2$, $\nu(2) = 4$. The membership antecedent is $\mathrm{Bit}(4,2)$ and is true; the conclusion is $\mathrm{Bit}(4,3)$ and is false.

Theorem 2. *Head 4 is not derivable from the remaining exact PA196 native rules, even with primitive equality reflexivity, symmetry and transitivity.*

5 A useful positive consequence of just head 4

This section uses head 4, head 5, the other logical heads and the gate, but never head 3. Vacuity for arbitrary formulas at a fresh label follows as in the upstream construction; that proof uses the gate, negation, implication and quantifier commutation, not either membership rewrite head.

Lemma 3 (Negative-membership specialization)**.** *For $r \neq x$ and $z \neq x$,*

$$\mathsf{U}(x, \mathsf{NM}(r, x)) \to \mathsf{NM}(r, z)$$

is derivable using head 4.

Proof. The equality witness is $\exists x_x \mathsf{E}(z, x)$. Head 4 gives $\mathsf{E}(z, x) \to (\mathsf{M}(r, z) \to \mathsf{M}(r, x))$. Under the assumption $\mathsf{M}(r, z)$, its vacuity at x and universal distribution give $\mathsf{U}(x, \mathsf{E}(z, x) \to \mathsf{M}(r, x))$. Combining this with the universal negative premise gives $\mathsf{U}(x, \mathsf{NE}(z, x))$, contradicting the witness. Discharge the assumption. □

Lemma 4 (Same-variable universal elimination)**.** *For every label x and formula code A,*

$$\mathsf{U}(x, A) \to A$$

is derivable without head 3.

Proof. Let c be the code of $\mathsf{N}A$. If $c = x$, use the code of $\mathsf{NNN}A$ instead; it is different and is propositionally equivalent to $\mathsf{N}A$. The gate at (x, x, c) gives a witness x_c with a point clause implying $\mathsf{M}(c, x)$ from $\mathsf{N}A$ and a separation clause implying $\mathsf{U}(x, \mathsf{M}(c, x) \to \mathsf{N}A)$. Assume $\mathsf{U}(x, A)$ and $\mathsf{N}A$. The clauses give $\mathsf{M}(c, x)$ and $\mathsf{U}(x, \mathsf{NM}(c, x))$.

Choose a label z fresh for all displayed formulas and distinct from $0, x, c$. Negative-membership specialization gives $\mathsf{NM}(c, z)$. Head 4 gives $\mathsf{E}(x, z) \to (\mathsf{M}(c, x) \to \mathsf{M}(c, z))$. Thus under the present assumptions $\mathsf{NE}(x, z)$ follows. Generalize at z; the assumptions are vacuous there. This contradicts head 5's $\exists x_z \mathsf{E}(x, z)$. Finally eliminate the separation witness at c, which is fresh in the desired conclusion, and discharge $\mathsf{N}A$ by classical logic. □

This result does not cross from a bound label x to a different free label y. Such a crossing is exactly where an attempted proof of the missing membership head needs an additional, unjustified substitution or alpha-conversion argument. The two counterinterpretations show that no completion of that route can produce the missing head from the unchanged remaining rules.

6 Limits of the conclusion

The countermodels exclude a specific reduction justification: "delete one head because the other heads already derive it." They do not prove that a machine searching the weaker calculus cannot still be equiconsistent with PA through a different interpretation. Nor do they establish any lower bound on Turing machine states. A new PA interpretation of a one-head calculus would be a separate mathematical construction.